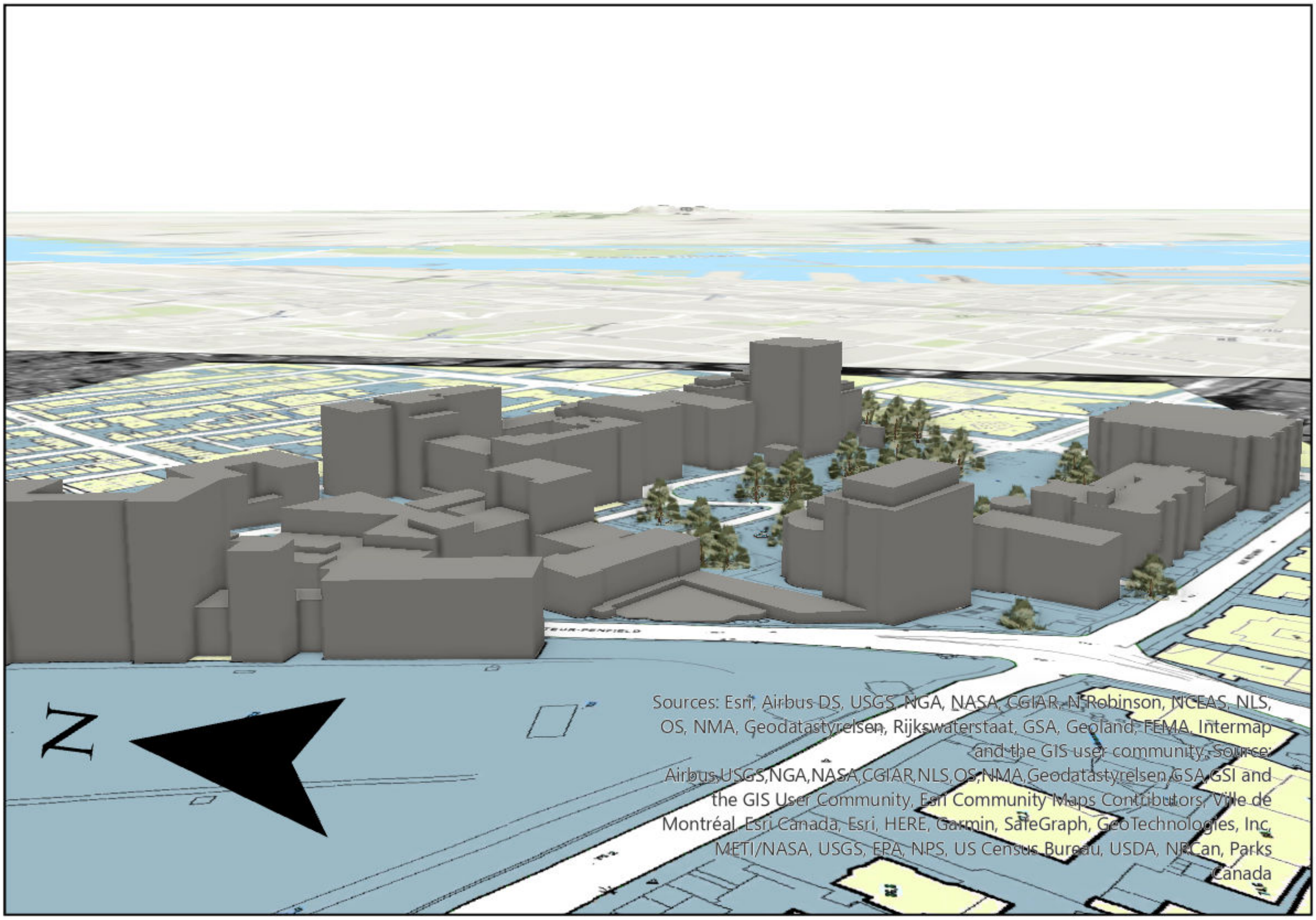
3-D Model of McGill Campus



0.01 0 0 0.01 0.02 0.02 0.03
Kilometers

Legend

- fountains
- buildings
- trees

Data Source: GEOG 44
Projected Coordinate System: NAD 1983 (NSRS2007)

Author: Grady Zappone
Date: 5 November 2023

This is a 3D scene depicting McGill University main campus' buildings. Some of the prominent buildings depicted in the scene include: The Arts Building, Leacock, Ferrier, and the McTavish reservoir building. The main road intersection displayed is McTavish St and Dr. Penfield Ave. Trees on campus and the Three Bares Fountain can also be seen.

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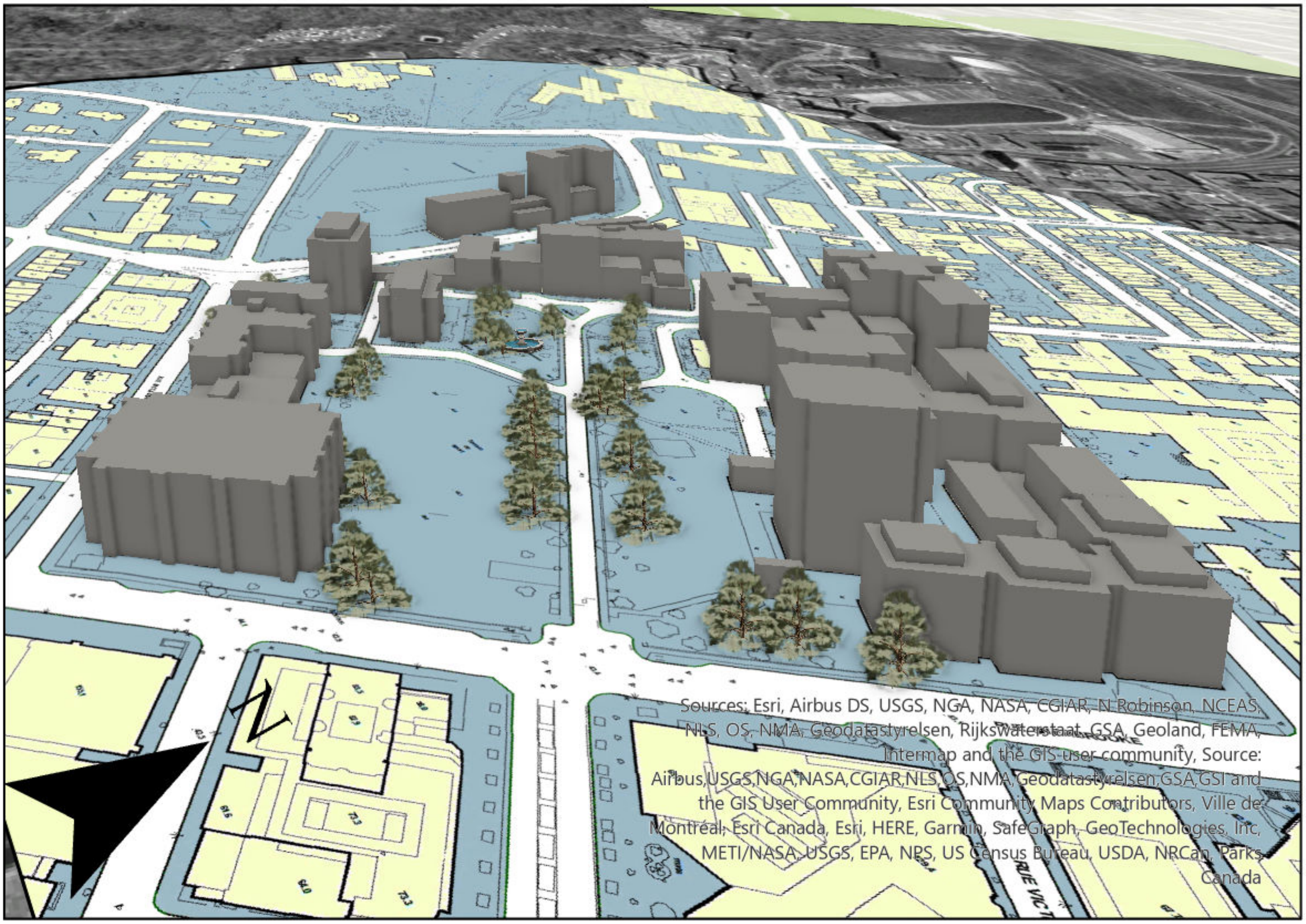

0.01 0.01 0 0.01 0.02 0.04 0.05
Kilometers

Legend

- fountains
- buildings
- trees

Data Source: GEOG 414
Projected Coordinate System: NAD 1983 (NSRS2007)

Author: Grady Zappone
Date: 5 November 2023

This is a 3D scene depicting McGill University main campus' buildings. Some of the prominent buildings depicted in the scene include: The McLennan-Redpath Complex, Burnside Hall, and the Otto Maass Chemistry Building. The main road displayed is Sherbrook St. W., and Mount Royal can be seen in the background. Trees on campus and the Three Bares Fountain can also be seen.

Sources of Error

Creating the 3D scene of McGill involved several steps, each with it's a potential for error and limitations that can impact the final accuracy of the model. One source of error is the geo-referencing process. The accuracy of the geo-referenced images (both the aerial photograph and the building footprints) is highly dependent on the precision with which control points were chosen and matched between the image and the base map. If the points were inaccurately placed or there were not enough points, the map may be misaligned, which can affect the spatial relationships in the scene. Interpolation errors are another concern. The process of creating a DEM from point elevation data using IDW interpolation assumes that the surface is smooth between known points and that point distribution is sufficient and evenly spread. This might not be the case, leading to a DEM that doesn't accurately reflect the terrain's nuances. When creating the 3D buildings, the precision of the digitized footprints and the accuracy of the height attributes are critical. If the building footprints are not traced accurately, or if the height data is incorrect or estimated, the resulting 3D buildings will not be a true representation of the real-world structures. Additionally, the rooftops of McGill buildings typically feature complex structures and varying elevations; however, the simplified 3D model represents these as uniform, flat surfaces at a single height, thus omitting the nuances of their actual architecture.